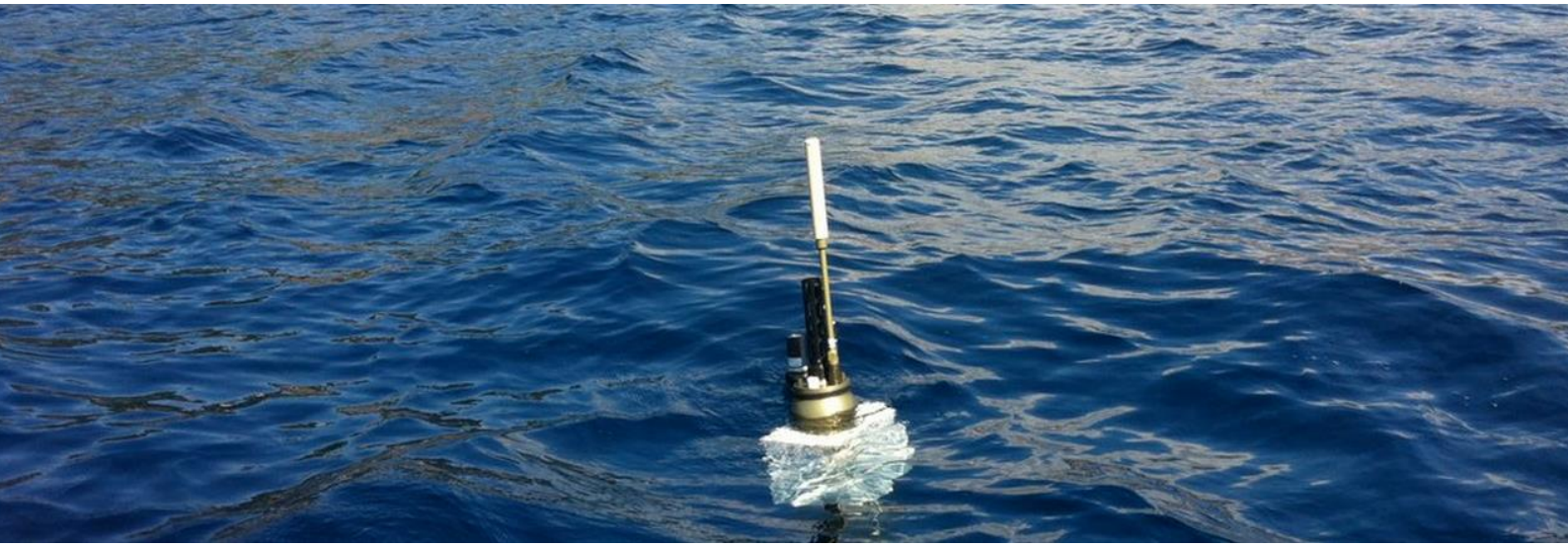
PFV2 Profiler - File management

PROFILING **F**LOAT **V**ECTOR **2**



NTxxxxxx_PFV2_File_Management_rev1.docx
Revision 1 (2024-04-09)

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1. Revision history

Revision	Release date	Notes	Author
1.0	2024-01	Original	D. NOGRE

1.1 Firmware upgrade

The following are the main changes to embedded software:

PFV2	Notes

2. File system

Float management is based on a specific file system developed by nke. This file system allows information and data to be stored when the float is turned off, together with the mission configuration.

These files can be accessed by an SFTP client via the WiFi connection.

2.1.Memory organisation

The memory space is divided into different areas intended for different uses. In particular, the following spaces can be found:

- Reserved: restricted system used to store file manipulation and management information
- Working: used for temporary storage (awaiting transmission).
- Storage: used to expand the temporary storage capacity of the workspace.

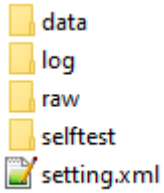
2.2.File types

The system uses various file types, as needed:

- Configuration file for mission and sensor settings
- Sensor data files (raw, processed and transmitted data)
- Technical data files
- Execution trace files (mission progress monitoring)
- Specific files (restricted system use)

2.3.Main memory – Working memory

The working memory (fast access) is itself divided into 4 directories:



- Folder for storing the current processed data
- Folder for storing the execution trace files
- Folder for storing the current raw data
- Folder for storing the result of the last selftest
- Configuration file embedding the mission and sensors descriptions

2.4.Extended memory – Storage memory

The use of an additional memory (slow access) increases the system's storage. In this case, files awaiting of transmission and to be moved from the working memory to the extended memory.

archive - Folder for storing the data already transmitted
data - Folder for storing the data in waiting of transmission
raw - Folder for storing the raw data of all previous missions and cycles

2.5.GZ compression

GZ compression is applied to all files. This compression is an option that can be deactivated if necessary. However, it is recommended that it is retained to improve transmission performance. The fact that there is less data to transmit saves energy and reduces Iridium's costs.

The principle of GZ compression involves a single file per compression. This file is then identified by keeping its name and adding the ".gz" extension.

If one of these files is fragmented, the cut can be made at any point in the file. The name extension is then formatted as ".gz.xxx", where "xxx" is the file assembly number.

To keep the filename size as small as possible, the file is renamed during compression. This name is replaced by a unique hexadecimal transmission number. Each file will therefore take a different number which will be incremented over transmissions. An optimization only possible since the original name of the file is integrated into the compression.



2.6.SBD decoding

Iridium SBD transmission is based on sending a message with a maximum size of 360 bytes (9602/9603 modem). If a larger file is sent, fragmentation is necessary. To enable the file to be reconstituted after transmission, a header is included at the beginning of this buffer for each transmission.

2.6.1. Header

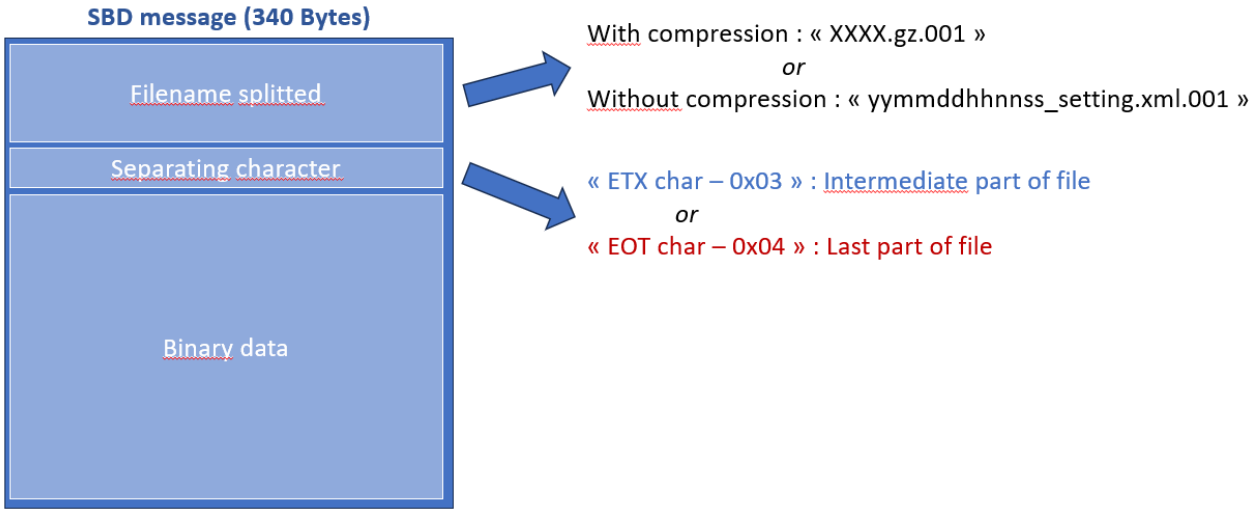
This header contains the name of the fragmented file and a separator (specific character). The purpose of the separator is to define the boundary between the header and the file's data, and to identify the last SBD of the fragmented file.

2.6.2. Separating character

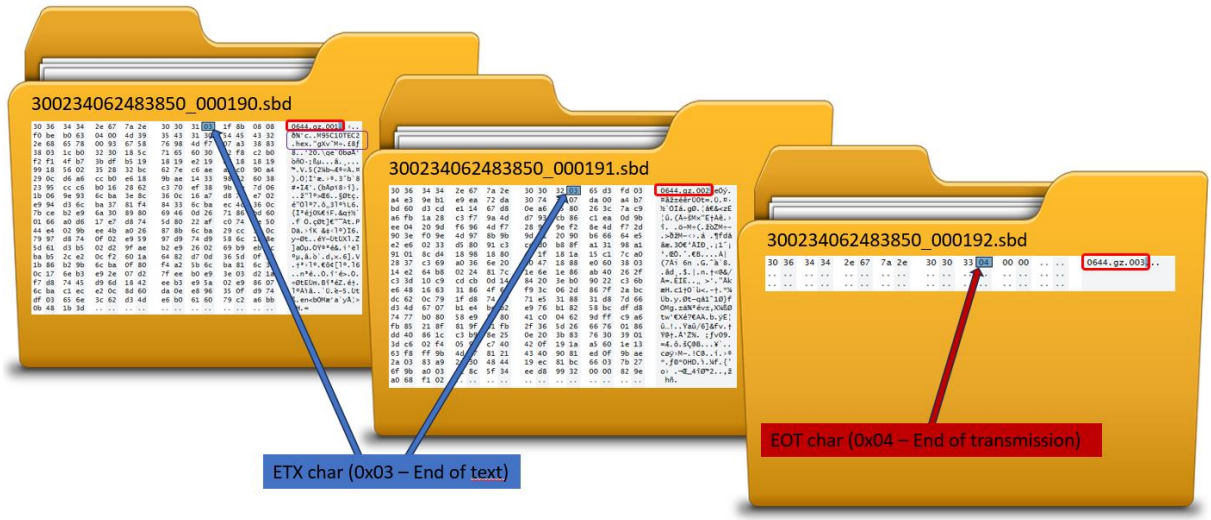
The character can only take 2 values:

- "ETX" char (0x03) meaning that it is an intermediate part of the file
- "EOT" char (0x04) meaning this is the last part of the fragmented file

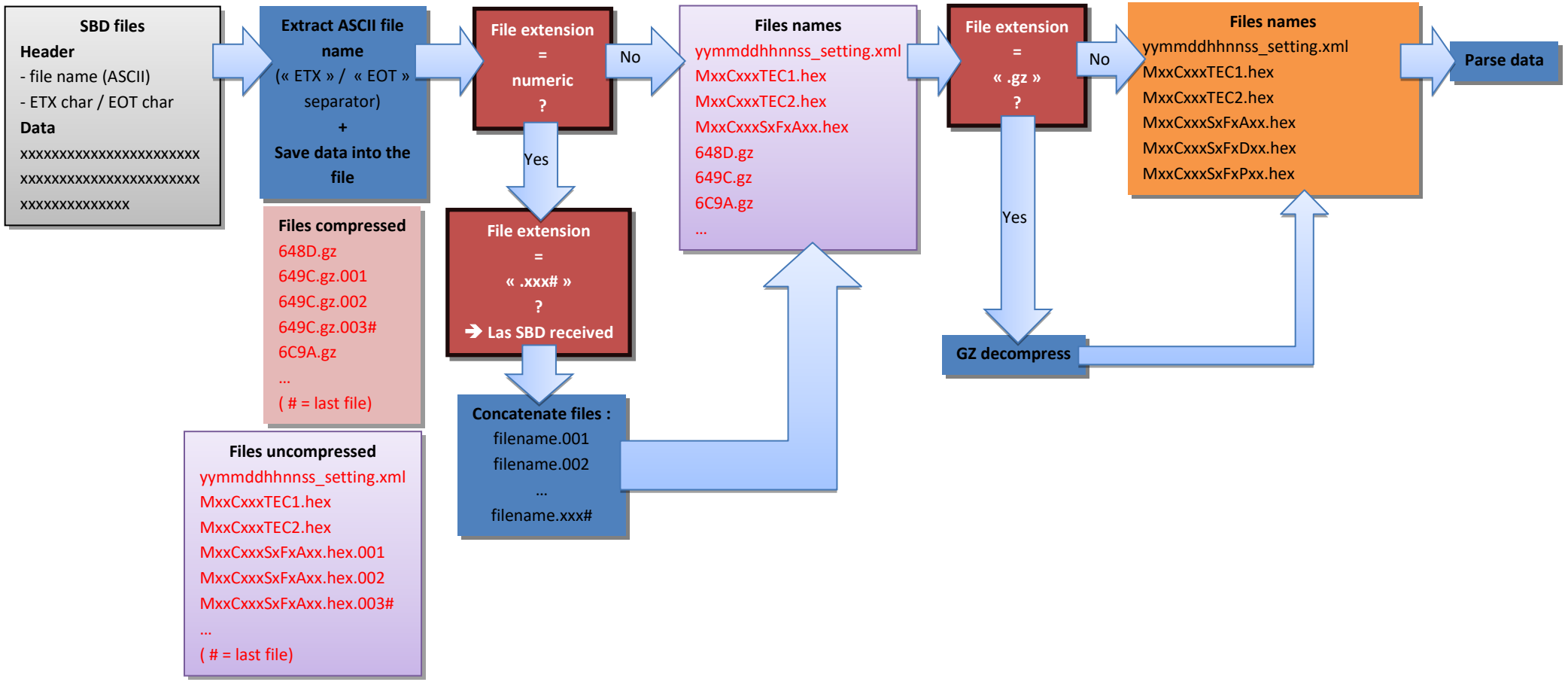
SBD content:



SBD encoding illustration:



2.6.3. Principle



P.S.: Python routines available, please contact nke Instrumentation.

3. Configuration files

The mission configuration is saved as a XML file organised by nodes and tags.

3.1 Local configuration via WiFi

During the mission configuration phase performed locally via WiFi, this file can be modified by the user. It can be manually uploaded in full using a SFTP client or partially modified with a command interpreter via a SSH client.

3.2 Remote configuration via remote commands

When the mission is in progress, the file can be modified by sending remote commands via the satellite communication system (SBD, RUDICS...).

3.3 Transmission of the configuration file

The system transmits the current configuration file at various stages of the mission:

- At the start of the mission (initial configuration)
- Upon a change of configuration following the application of a remote command (optional for an Iridium SBD transmission)

The various files generated are named using the format:

Iridium SBD : "yymmddhhnnss_SBD_cmd.ini"

- with "yymmddhhnnss": Date/Time of the float (12 characters)

Iridium RUDICS : “yymmddhhnnss_RUDICS_cmd.txt”

- with "yymmddhhnnss": Date/Time of the float (12 characters)

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<SETTING P0="347F8EFF">
<PRODUCT P0="ARVOR" P1="IRIDIUM SBD"/>
<MISSION P0="0">
        <PRESSURE_DETECTION P0="300" P1="10" P2="90"/>
        <INITIALIZATION P0="0"/>
        <LOOP01 P0="1">
                <CYCLE01 P0="1" P1="0" P2="">
                        <TRANSMISSION01 P0="false"/>
                        <BUOYANCY_RED P0="1"/>
                        <DESC_PARK P0="1000" P1="2.5"/>
                        <PARK_DRIFT P0="0" P1="50" P2="30"/>
                        <DESC_PROF P0="2000" P1="2.5"/>
                        <PROF_DRIFT P0="0" P1="50" P2="30"/>
                        <ASCENT P0="0" P1="9"/>
                        <EMERGENCE/>
                        <IN_AIR P0="0" P1="300"/>
                        <TRANSMISSION02/>
                        <END_OF_CYCLE P0="0"/>
                </CYCLE01>
                <CYCLE02 P0="999" P1="864000" P2="12:00:00">
                        <TRANSMISSION01 P0="true"/>
                        <BUOYANCY_RED P0="1"/>
                        <DESC_PARK P0="1000" P1="2.5"/>
                        <PARK_DRIFT P0="0" P1="50" P2="30"/>
                        <DESC_PROF P0="2000" P1="2.5"/>
                        <PROF_DRIFT P0="0" P1="50" P2="30"/>
                        <ASCENT P0="0" P1="9"/>
                        <EMERGENCE/>
                        <IN_AIR P0="2" P1="300"/>
                        <TRANSMISSION02/>
                        <END_OF_CYCLE P0="0"/>
                </CYCLE02>
        </LOOP01>
        <END_OF_LIFE P0="3600"/>
</MISSION>
<TECHNICAL>
        <BUOYANCY_RED P0="1" P1="7" P2="60" P3="7"/>
        <DESCENT P0="30">
                <MIN P0="2.5" P1="1" P2="300" P3="11"/>
                <MAX P0="9" P1="3" P2="100" P3="20"/>
        </DESCENT>
        <DRIFT P0="1800"/>
…
```

4. Technical files

The various pieces of technical information associated with a profile are saved in a binary file organised by sections.

4.1.Transmission of technical information

The system transmits technical files at various stages of the mission:

- During system verification prior to the mission (self-test)
- During the mission at the end of each cycle
- At end of life

4.1.1. Self-test files

The files generated are named using the format "yymmddhhnnss_selfest.hex":

- with "yymmddhhnnss": Date/Time of the float (12 characters)

4.1.2. Mission files

The files generated are named using the format "MxCxTECx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “TECx” : with “x” corresponding the number of the transmission’s session in the cycle

4.1.3. End-of-life files

The various files generated are named using the format "MxCxEOL.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle

4.2. Technical information available

The data is composed of a serial of events. The chronology of these events describes how the cycle unfolded.

An event is defined as follows:

- Type of event (2 Bytes)
- Data event (variable size depending on the type of event)

4.2.1. Product information events

Type number (2 Bytes)	Data event	Format
0	**Product infos**	
	Float ID	CHAR [32]
	Firmware version	
	- Type	CHAR [6]
	- Version	U_INT8
	- Subversion	U_INT8
	- Release	U_INT16
	Firmware checksum	U_INT32
1	**Mission infos**	
	Number of the current mission	U_INT16
	Number of the current cyle	U_INT16
	Cycle's loop setting index	U_INT8
	Cycle setting index	U_INT8
2	**Life infos**	
	Total profile realized	U_INT16
	Total distance profiled (dbar)	IEEE_FLOAT32
	Total duration at sea (s)	U_INT132

4.2.2. Phases events

Type number (2 Bytes)	Data event	Format
1000	**Buoyancy reduction beginning**	
	Date/Time (in second since 2000)	U_INT32
1001	**Buoyancy reduction end**	
	Date/Time (in second since 2000)	U_INT32
	EV lump sum volume (1/10 cm3)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
1002	**Buoyancy inversion beginning**	
	Date/Time (in second since 2000)	U_INT32
1003	**Buoyancy inversion end**	



	Date/Time (in second since 2000)	U_INT32
	External pressure offset (cbar)	U_INT16
	External pressure min (cbar)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
1004	**Descent to parking beginning**	
	Date/Time (in second since 2000)	U_INT32
1005	**Decent to parking end**	
	Date/Time (in second since 2000)	U_INT32
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	External pressure max (cbar)	U_INT16
1006	**Descent to profile beginning**	
	Date/Time (in second since 2000)	U_INT32
1007	**Descent to profile end**	
	Date/Time (in second since 2000)	U_INT32
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	External pressure max (cbar)	U_INT16
1008	**Parking drift beginning**	
	Date/Time (in second since 2000)	U_INT32
1009	**Parking drift end**	
	Date/Time (in second since 2000)	U_INT32
	External pressure min (cbar)	U_INT16
	External pressure max (cbar)	U_INT16
	Stabilized entrance number	U_INT8
	Unstabilized entrance number	U_INT8
	Repositioning number	U_INT8
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8
1010	**Profile drift beginning**	
	Date/Time (in second since 2000)	U_INT32
1011	**Profile drift end**	
	Date/Time (in second since 2000)	U_INT32
	Ext. pressure min (cbar)	U_INT16
	Ext. pressure max (cbar)	U_INT16
	Stabilized entrance number	U_INT8
	Unstabilized entrance number	U_INT8
	Repositioning number	U_INT8
	EV transferred volume (1/10 cm3)	U_INT16
	EV action number	U_INT8
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8



1012	**Ascent beginning**	
	Date/Time (in second since 2000)	U_INT32
1013	**Ascent end**	
	Date/Time (in second since 2000)	U_INT32
	PUMP transferred volume (1/10 cm3)	U_INT16
	PUMP action number	U_INT8
1014	**Emergence**	
	Date/Time (in second since 2000)	U_INT32
1015	**In air**	
	Date/Time (in second since 2000)	U_INT32
1016	**Transmission**	
	Date/Time (in second since 2000)	U_INT32
1017	**Emergency ascent beginning**	
	Date/Time (in second since 2000)	U_INT32
	External pressure max (cbar)	U_INT16
1018	**Emergency ascent end**	
	Date/Time (in second since 2000)	U_INT32
	PUMP transferred volume (1/10 cm3)	U_INT16
1019	**End of life**	
	Date/Time (in second since 2000)	U_INT32
	Last date/time of end-of-life detection (in second since 2000)	U_INT32
	End of life source	U_INT32
	- BIT0: Programmed	
	- BIT1: Startup failure	
	- BIT2: Self-test failure	
	- BIT3: Settings failure	
	- BIT4: CTD settings failure	
	- BIT5: Request by downlink	
	- BIT6: Ext. pressure failure	
	- BIT7: Alarm on ext. pressure high	
	- BIT8: Alarm on ext. pressure error	
	- BIT9: Alarm on descent speed high	
	- BIT10: Alarm on battery voltage low	
	- BIT11: Alarm on insulation voltage	
	- BIT12: Alarm on RTC	
1020	**Pressure detection beginning**	
	Date/Time (in second since 2000)	U_INT32
1021	**Pressure detection end**	
	Date/Time (in second since 2000)	U_INT32
	Date/Time of external pressure detection	U_INT32
	External pressure max (dbar)	U_INT16

4.2.3. Security events

Type number (2 Bytes)	Data event	Format
2000	**Grounding**	
	Date/Time (in second since 2000)	U_INT32
	External Pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
2001	**Hanging**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
2002	**Braking**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	Speed (mm/s)	U_INT16
2003	**Emergency braking**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	Speed (mm/s)	U_INT16
2004	**ISA detection filtering**	
	Date/Time (in second since 2000)	U_INT32
	Counter of detection	U_INT8
	Number max of detection before confirming ice's presence	U_INT8
2005	**ISA result**	
	Date/Time (in second since 2000)	U_INT32
	Median temperature (m°C)	INT16
2006	**Ice detection**	
	Date/Time (in second since 2000)	U_INT32
	Type of detection	U_INT16
	- BIT0 : ISA detection	
	- BIT1 : Hanging detection	
	- BIT2 : Satellite mask	
	Emergence prohibited until date/time (in seconde since 2000)	U_INT32
	Emergence will be forced at date/time (in second since 2000)	U_INT32
2007	**Deployment mode**	
	Mode (0: not armed, 1: armed, 2: armed in pressure detection mode)	U_INT8
2008	**New setpoint**	
	Date/Time (in second since 2000)	U_INT32
	Previous setpoint (cbar)	U_INT16
	New setpoint (cbar)	U_INT16
2009	**Empty ballast detection**	
	Date/Time (in second since 2000)	U_INT32
	Detection flag	U_INT8

4.2.4. Alarms events

Type number (2 Bytes)	Data event	Format
3000	**Alarm without value**	
	Date/Time (in second since 2000)	U_INT32
	Alarm type	U_INT16
3001	**Alarm with value**	
	Date/Time (in second since 2000)	U_INT32
	Alarm type	U_INT16
	Value	IEEE_FLOAT32
	Threshold	IEEE_FLOAT32
3002	**SBE41 limit values**	
	Pressure: Number of samples over limits	U_INT8
	Pressure: Min value expected	IEEE_FLOAT32
	Pressure: Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature: Min value expected	IEEE_FLOAT32
	Temperature: Max value expected	IEEE_FLOAT32
	Salinity: Number of samples over limits	U_INT8
	Salinity: Min value expected	IEEE_FLOAT32
	Salinity: Max value expected	IEEE_FLOAT32
3003	**Aanderaa4330 limit values**	
	C1RPhase: Number of samples over limits	U_INT8
	C1RPhase: Min value expected	IEEE_FLOAT32
	C1RPhase: Max value expected	IEEE_FLOAT32
	C2RPhase: Number of samples over limits	U_INT8
	C2RPhase: Min value expected	IEEE_FLOAT32
	C2RPhase: Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature: Min value expected	IEEE_FLOAT32
	Temperature : Max value expected	IEEE_FLOAT32
3004	**RBRargo limit values**	
	Pressure: Number of samples over limits	U_INT8
	Pressure : Min value expected	IEEE_FLOAT32
	Pressure : Max value expected	IEEE_FLOAT32
	Temperature: Number of samples over limits	U_INT8
	Temperature : Min value expected	IEEE_FLOAT32
	Temperature : Max value expected	IEEE_FLOAT32
	Salinity: Number of samples over limits	U_INT8
	Salinity : Min value expected	IEEE_FLOAT32
	Salinity : Max value expected	IEEE_FLOAT32
	Coils temperature: Number of samples over limits	U_INT8
	Coils temperature: Min value expected	IEEE_FLOAT32

	Coils temperature: Max value expected	IEEE_FLOAT32
	Salinity corrected: Number of samples over limits	U_INT8
	Salinity corrected: Min value expected	IEEE_FLOAT32
	Salinity corrected: Max value expected	IEEE_FLOAT32
3005	**Sensor error**	
	Number of the sensor	U_INT8
	Number of invalid data	U_INT8
3006	**Invalid frame from sensor**	
	Date/Time (in second since 2000)	U_INT32
	Number of the sensor	U_INT8
	Frame size	U_INT16
	Sensor frame	CHAR ["Frame size"]

4.2.5. Float infos events

Type number (2 Bytes)	Data event	Format
4000	**Battery level**	
	Date/Time (in second since 2000)	U_INT32
	Voltage (mV)	U_INT16
4001	**Battery level at pressure max**	
	Date/Time (in second since 2000)	U_INT32
	Voltage (mV)	U_INT16
	External pressure (cbar)	U_INT16
4002	**Servitude measurement**	
	Date/Time (in second since 2000)	U_INT32
	Internal pressure (mbar)	U_INT16
	Internal temperature (m°C)	INT16 + offset (20000m°C)
	Internal relative humidity (%RH)	U_INT8
4003	**Internal pressure compensed**	
	Internal pressure (mbar)	U_INT16
4005	**Reset**	
	Date/Time (in second since 2000)	U_INT32
	Source of the reset	U_INT8
	- BIT0 : Firewall	
	- BIT1 : Option byte loader	
	- BIT2 : PIN reset	
	- BIT3 : BOR reset	
	- BIT4 : software	
	- BIT5 : Independent watchdog	
	- BIT6 : Window watchdog	
	- BIT7 : Low-power reset	
4006	**Memory disk**	



	Disk number (0: Flash, 1: SDcard)	U_INT8
	Disk occupancy (resolution: 0.5%)	U_INT8

4.2.6. Sensors state events

Type number (2 Bytes)	Data event	Format
5000	**SBE41 subsurface point**	
	CTD pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
5001	**SBE41 offset pressure**	
	Offset value	IEEE_FLOAT32
5002	**RBRargo3 offset pressure**	
	Offset value	IEEE_FLOAT32

4.2.7. Selftest events

Type number (2 Bytes)	Data event	Format
6000	**Self-test result**	
	FRAM memory result flag (0: error, 1: ok)	U_INT8
	FLASH memory result flag (0: error, 1: ok)	U_INT8
	SDcard state (0: error, 1: ok, 2: not present)	U_INT8
	XML settings result flag (0: error, 1: ok)	U_INT8
	RTC result flag (0: error, 1: ok)	U_INT8
	RTC date/Time (in second since 2000)	U_INT32
	Insulation result flag (0: error, 1: ok)	U_INT8
	Insulation value	U_INT16
	Battery result flag (0: error, 1: ok)	U_INT8
	On load battery voltage (mV)	U_INT16
	EV result flag (0: error, 1: ok)	U_INT8
	PUMP result flag (0: error, 1: ok)	U_INT8
	Empty ballast sensor result flag (0: error, 1: ok)	U_INT8
	Max buoyancy pumping duration (s)	U_INT16
	Internal pressure result flag (0: error, 1: ok)	U_INT8
	Internal pressure (mbar)	IEEE_FLOAT32
	Internal pressure compensed (mbar)	IEEE_FLOAT32
	Internal temperature result flag (0: error, 1: ok)	U_INT8
	Internal temperature (°C)	IEEE_FLOAT32
	Internal relative humidity result flag (0: error, 1: ok)	U_INT8
	Internal relative humidity (%)	IEEE_FLOAT32
	GNSS result flag (0: error, 1: ok)	U_INT8
	Modem result flag (0: error, 1: ok)	U_INT8



	Dummy buffer (allowing future evolutions)	U_INT8 * 32
6001	**SBE41 self-test result**	
	Fast pressure result flag (0: error, 1: ok)	U_INT8
	Fast pressure (dbar)	IEEE_FLOAT32
	CTD measurement result flag (0: error, 1: ok)	U_INT8
	CTD Pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
6002	**Aanderaa4330 self-test result**	
	Measurement result flag (0: error, 1: ok)	U_INT8
	Serial number	U_INT16
6003	**RBRargo3 self-test result**	
	Fast pressure result flag (0: error, 1: ok)	U_INT8
	Fast pressure (dbar)	IEEE_FLOAT32
	CTD measurement result flag (0: error, 1: ok)	U_INT8
	CTD pressure (dbar)	IEEE_FLOAT32
	CTD temperature (°C)	IEEE_FLOAT32
	CTD salinity (PSU)	IEEE_FLOAT32
	CTD coils temperature (°C)	IEEE_FLOAT32
	CTD corrected salinity (PSU)	IEEE_FLOAT32
	Sensor's serial number	U_INT32

4.2.8. Transmission events

Type number (2 Bytes)	Data event	Format
7000	**Geographic location**	
	Date/Time (in second since 2000)	U_INT32
	Latitude (°)	IEEE_FLOAT32
	Longitude (°)	IEEE_FLOAT32
	Number of satellite	U_INT8
	Valid fix	U_INT8
	Clock drift (s) **(*)**	IEEE_FLOAT32
	Session duration (s)	U_INT16
7001	**SBD Iridium (*)**	
	Date/Time (in second since 2000)	U_INT32
	Session duration (s)	U_INT16
	SBDI success counter	U_INT16
	SBDI total counter	U_INT16
	Remote command accepted counter	U_INT8
	Remote command refused counter	U_INT8
	Remote command unkown counter	U_INT8
	Received files number	U_INT16
	Received size (Byte)	IEEE_FLOAT32
	Transmitted files number	U_INT16

	Transmitted size (Byte)	IEEE_FLOAT32
	Files number in waiting of transmission	U_INT16
7002	**RUDICS Iridium (*)**	
	Date/Time (in second since 2000)	U_INT32
	Session duration (s)	U_INT16
	Session number	U_INT8
	Remote command accepted counter	U_INT8
	Remote command refused counter	U_INT8
	Remote command unkown counter	U_INT8
	Received files number	U_INT16
	Received size (Byte)	IEEE_FLOAT32
	Transmitted files number	U_INT16
	Transmitted size (Byte)	IEEE_FLOAT32
	Files number in waiting of transmission	U_INT16

4.2.9. Navigation events

Type number (2 Bytes)	Data event	Format
8000	**First stabilization (on descent)**	
	Date/Time (in second since 2000)	U_INT32
	Stabilization state	U_INT8
	External pressure (cbar)	U_INT16
8001	**Pressure monitoring**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
8002	**Pump action**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	PUMP transferred volume (1/10 cm3)	U_INT16
8003	**Ev action**	
	Date/Time (in second since 2000)	U_INT32
	External pressure (cbar)	U_INT16
	EV transferred volume (1/10 cm3)	U_INT16

4.2.10. Treatment events

Type number (2 Bytes)	Data event	Format
9000	**Treated data in profile**	
	Date/Time (in second since 2000)	U_INT32
	Number of the sensor	U_INT8
	Number of the phase	U_INT8
	Zone 1 Number of samples	U_INT16

		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 2	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 3	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 4	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
	Zone 5	Number of samples	U_INT16
		Treatment type (0x00 : raw, 0x80 : raw decimated, 0x81 : average decimated)	U_INT8
9001	**Treated data in drift**		
	Date/Time (in second since 2000)		U_INT32
	Number of the sensor		U_INT8
	Number of the phase		U_INT8
	Number of samples		U_INT16

(*) Clock drift = Float time - GPS time. The offset is reset to 0 after every GPS retiming.

(*) Information from the previous transmission session.

Example of parsing:

Previous Iridium session infos -> 2023/12/11 02:58:29
Session duration : 462 s
SBDI success : 21 / 21
Command accepted : 0
Command refused : 0
Command unknown : 0
SBD received : 0 (Size: 0 B)
SBD transmitted : 5 (Size: 6645 B)
Files in waiting : 0
Product life -> Total profiles realized: 255 Total distance profiled: 34997 dbar Total duration at sea: 35 days, 5 hours
MISSION 95 - CYCLE 21 (LOOP01 - CYCLE01)
BUOYANCY REDUCTION - 2023/12/11 02:58:32
Internal measures -> Pressure: 1009 mbar
Temperature: 20.158 °C
Humidity: 51 %RH
Battery voltage: 10.669 V
SBE41 - Pressure offset: 0.000 cbar
Ev forfaitaire volume: 0.0 cm3
Ev transfered volume: 686.1 cm3
Ev action number: 99
BUOYANCY REDUCTION END - 2023/12/11 03:11:24

DESCENT TO PARKING - 2023/12/11 03:11:25
Pressure infos -> 2023/12/11 03:16:30 Pressure: 22.4 dbar
Pressure infos -> 2023/12/11 03:21:30 Pressure: 33.2 dbar
Pressure infos -> 2023/12/11 03:26:32 Pressure: 41.2 dbar
Pressure infos -> 2023/12/11 03:31:32 Pressure: 47.3 dbar
Pressure infos -> 2023/12/11 03:36:32 Pressure: 51.7 dbar
Pressure infos -> 2023/12/11 03:41:32 Pressure: 55.1 dbar
1st stabilization -> 2023/12/11 03:41:32 Pressure: 55.1 dbar

Ev action -> 2023/12/11 03:41:33 Pressure: 55.1 dbar Transfered volume: 0.3 cm3
Pressure infos -> 2023/12/11 03:46:32 Pressure: 57.9 dbar
Ev action -> 2023/12/11 03:46:34 Pressure: 57.9 dbar Transfered volume: 11.0 cm3
Pressure infos -> 2023/12/11 03:51:32 Pressure: 74.0 dbar
Pressure infos -> 2023/12/11 03:56:32 Pressure: 86.1 dbar
Pressure infos -> 2023/12/11 04:01:32 Pressure: 95.2 dbar
Pressure infos -> 2023/12/11 04:06:34 Pressure: 102.0 dbar
Pressure infos -> 2023/12/11 04:11:34 Pressure: 107.0 dbar
Pressure infos -> 2023/12/11 04:16:34 Pressure: 110.8 dbar
Pressure infos -> 2023/12/11 04:21:34 Pressure: 113.6 dbar
Ev action -> 2023/12/11 04:21:36 Pressure: 113.6 dbar Transfered volume: 11.0 cm3
Pressure infos -> 2023/12/11 04:26:34 Pressure: 133.9 dbar
Pressure infos -> 2023/12/11 04:31:34 Pressure: 149.0 dbar
Pressure infos -> 2023/12/11 04:36:34 Pressure: 160.4 dbar
Pressure infos -> 2023/12/11 04:41:36 Pressure: 168.9 dbar
Pressure infos -> 2023/12/11 04:46:36 Pressure: 175.2 dbar
Pressure infos -> 2023/12/11 04:51:36 Pressure: 179.9 dbar
Pressure infos -> 2023/12/11 04:56:36 Pressure: 183.5 dbar
Pressure infos -> 2023/12/11 05:01:36 Pressure: 186.1 dbar
Ev action -> 2023/12/11 05:01:37 Pressure: 186.1 dbar Transfered volume: 6.3 cm3
Pressure infos -> 2023/12/11 05:06:36 Pressure: 204.7 dbar
Pressure infos -> 2023/12/11 05:11:36 Pressure: 218.7 dbar
Pressure infos -> 2023/12/11 05:16:36 Pressure: 229.2 dbar
Pressure infos -> 2023/12/11 05:21:38 Pressure: 237.0 dbar
Pressure infos -> 2023/12/11 05:26:38 Pressure: 242.8 dbar
Pressure infos -> 2023/12/11 05:31:38 Pressure: 247.2 dbar
Pressure infos -> 2023/12/11 05:36:38 Pressure: 250.4 dbar
Ev action -> 2023/12/11 05:36:39 Pressure: 250.4 dbar Transfered volume: 2.1 cm3
Pressure infos -> 2023/12/11 05:41:40 Pressure: 260.0 dbar
Pressure infos -> 2023/12/11 05:46:40 Pressure: 267.1 dbar
Pressure infos -> 2023/12/11 05:51:40 Pressure: 272.4 dbar
Pressure infos -> 2023/12/11 05:56:40 Pressure: 276.3 dbar
Pressure infos -> 2023/12/11 06:01:42 Pressure: 279.3 dbar
Ev transfered volume: 30.7 cm3
Ev action number: 5
Pressure max: 279.3 dbar
DESCENT TO PARKING END - 2023/12/11 06:01:43

PARKING DRIFT - 2023/12/11 06:01:43
Treated data infos -> Sensor: 1 Phase: DESC_TO_PARK Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
Zone 2: Sample number= 190 Treatment= AVERAGE DECIMATED
Zone 3: Sample number= 80 Treatment= AVERAGE DECIMATED
Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Treated data infos -> Sensor: 2 Phase: DESC_TO_PARK Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
Zone 2: Sample number= 240 Treatment= AVERAGE DECIMATED
Zone 3: Sample number= 30 Treatment= AVERAGE DECIMATED
Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED

Pressure min:	279.4 dbar
Pressure max:	279.4 dbar
Stabilized entrance number:	0
Unstabilized entrance number:	0
Repositioning number:	0
Ev transfered volume:	0.0 cm3
Ev action number:	0
Pump transfered vol.:	0.0 cm3
Pump action number:	0

PARKING DRIFT END - 2023/12/11 06:01:54

DESCENT TO PROFILE - 2023/12/11 06:01:55

Ev transfered volume:	0.0 cm3
Ev action number:	0

```
Pressure max:                       279.5 dbar
### DESCENT TO PROFILE END - 2023/12/11 06:02:05 ###

### PROFILE DRIFT - 2023/12/11 06:02:05 ###
Pressure min:                       279.6 dbar
Pressure max:                       279.6 dbar
Stabilized entrance number:         0
Unstabilized entrance number:       0
Repositioning number:               0
Ev transfered volume:               0.0 cm3
Ev action number:                   0
Pump transfered vol.:               0.0 cm3
Pump action number:                 0
### PROFILE DRIFT END - 2023/12/11 06:02:14 ###

### ASCENT - 2023/12/11 06:02:15 ###
Internal measures ->                Pressure:           1010 mbar
                                    Temperature:        20.748 °C
                                    Humidity:           51 %RH
Pump action -> 2023/12/11 06:02:34 Pressure: 279.7 dbar Transfered volume: 11.0 cm3
Battery voltage min:                9.498 V (279.7 dbar)
Pump action -> 2023/12/11 06:04:38 Pressure: 269.4 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:16:38 Pressure: 186.1 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:24:38 Pressure: 139.5 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:28:38 Pressure: 118.4 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:38:38 Pressure: 57.1 dbar Transfered volume: 11.0 cm3
Pump action -> 2023/12/11 06:44:40 Pressure: 22.1 dbar Transfered volume: 11.0 cm3
Pump transfered vol.:               77.0 cm3
Pump action number:                 7
### ASCENT END - 2023/12/11 06:46:31 ###

### EMERGENCE - 2023/12/11 06:46:35 ###
Pump action -> 2023/12/11 06:51:59 Pressure: 0.0 dbar Transfered volume: 420.0 cm3
### TRANSMISSION - 2023/12/11 06:52:04 ###
Memory disk usage -> Drive: FLASH   Space used: 26.5 %
Memory disk usage -> Drive: SDCARD  Space used: 0.0 %
Subsurface point ->                 Pressure:           5.192 dbar
                                    Temperature:        0.000 °C
                                    Salinity:           32.680 PSU
Treated data infos -> Sensor: 1 Phase: ASCENT   Zone 1: Sample number= 11 Treatment= AVERAGE DECIMATED
                                                Zone 2: Sample number= 190 Treatment= AVERAGE DECIMATED
                                                Zone 3: Sample number= 80 Treatment= AVERAGE DECIMATED
                                                Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
                                                Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Treated data infos -> Sensor: 2 Phase: ASCENT   Zone 1: Sample number= 7 Treatment= AVERAGE DECIMATED
                                                Zone 2: Sample number= 224 Treatment= AVERAGE DECIMATED
                                                Zone 3: Sample number= 24 Treatment= AVERAGE DECIMATED
                                                Zone 4: Sample number= 0 Treatment= AVERAGE DECIMATED
                                                Zone 5: Sample number= 0 Treatment= AVERAGE DECIMATED
Geographic location ->              UTC :               2023/12/11 06:53:05
                                    Latitude :          47.7978287 °
                                    Longitude :         -3.2856851 °
                                    Satellite number:   3
                                    Valid fix :         1
                                    Clock drift :       0.424 s
                                    Session duration:   57 s
```

5. Sensor data files

The system generates sensor data files in different formats depending on the sensors.

5.1.Files format

5.1.1. Iridium SBD format

The files generated are named using the format "MxCxSxFxYx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “Sx” : with “x” corresponding the number of the sensor
- “Fx” : with “x” corresponding the type of data’s format (different for each sensor)
- “Yx” : with “Y” corresponding the phase
    - “D”: Descent
    - “P”: Parking drift
    - “A”: Ascent
    - “S”: Surface

  And with “x” being the datation frequency where:
    - 0: No date
    - 1: One date on each sensor’s data
    - 15: One date every 15 sensor’s data

5.1.2. Iridium RUDICS format

The files generated are named using the format "MxCxSxFx.hex":

- "Mx" : with “x” corresponding the number of the mission
- "Cx” : with “x” corresponding the number of the cycle
- “Fx” : with “x” corresponding the type of data’s format (different for each sensor)

Unlike the SBD format, each sample is dated.

5.1.2.1. Phase information format

The phase informations are encoded by setting the date to 0xFFFFFFF.

Value	Format	Calculation
Date/Time	U_INT32	Set to 0xFFFFFFFF
Phase	U_INT8	- 0: Descent to parking - 1: Parking drift - 2: Descent to profile - 3: Profile drift - 4: Ascent - 5: Surface
Zone of acquisition	U_INT8	From 1 to 5
Treatment type	U_INT8	- 0x00 : raw - 0x80 : raw decimated - 0x81 : average decimated

The values corresponding to the acquisition zone and the type of treatment are set to 0xFF for the phases “Parking drit”, “Profile drift” and “Surface” phases.

5.2. Data format

5.2.1. SBE41 - “S0”

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
F1	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n

5.2.2. Aanderaa 4330 - “S1”

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	C1RPhase (°)	U_INT16	40	500	C1(°) = n/500 – 40
	C2RPhase (°)	U_INT16	40	500	C2(°) = n/500 – 40
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
F1	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	C1RPhase (°)	IEEE_FLOAT32			C1(°) = n
	C2RPhase (°)	IEEE_FLOAT32			C2(°) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n

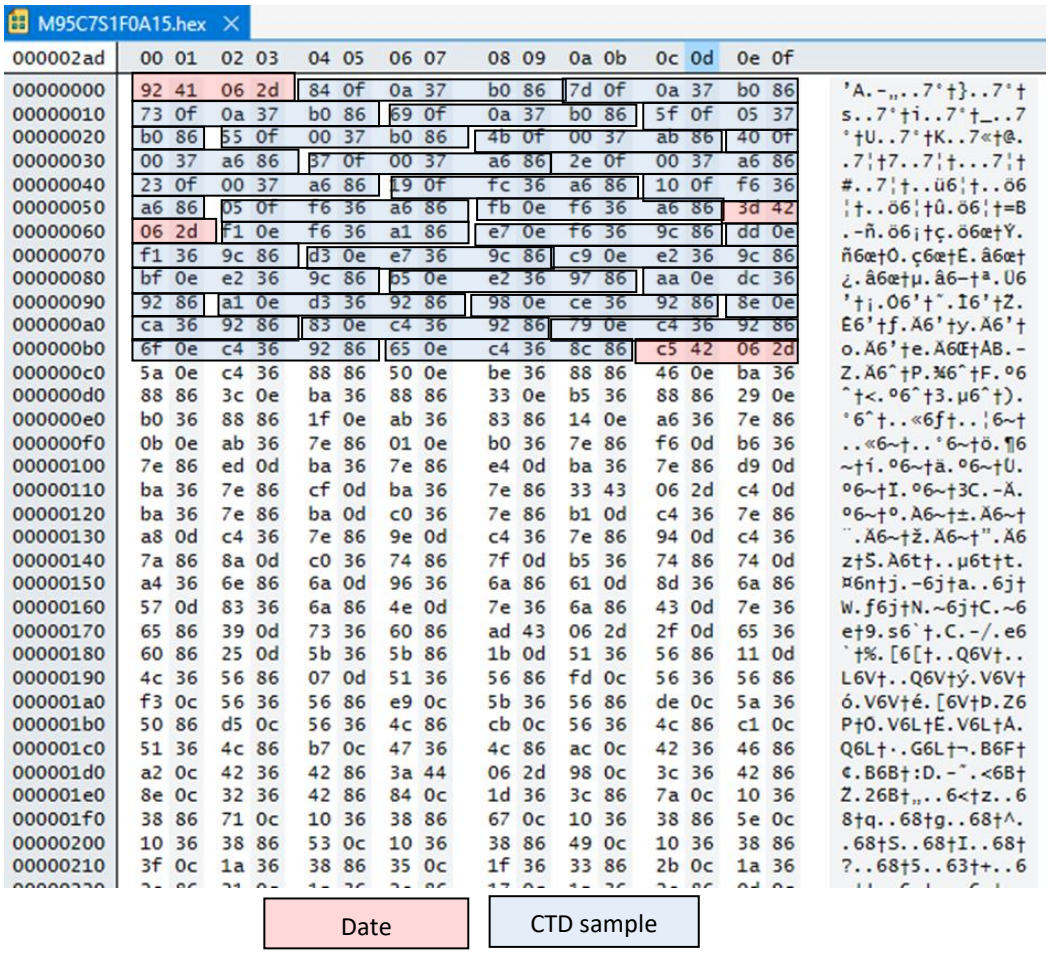
5.2.3. RBRargo3 – “S2”

Format	Value	Format	Offset	Slope	Calculation
F0	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000

F1	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Coils temperature (°C)	U_INT16	10	1000	Tc(°C) = n/1000 - 10
F2	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F3	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity (PSU)	U_INT16	0	1000	S(PSU) = n/1000
	Coils temperature (°C)	U_INT16	10	1000	Tc(°C) = n/1000 - 10
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F4	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	U_INT16	100	10	P(dbar) = n/10 - 100
	Temperature (°C)	U_INT16	10	1000	T(°C) = n/1000 - 10
	Salinity corrected (PSU)	U_INT16	0	1000	Sc(PSU) = n/1000
F5	Date/Time (in second since 2000)	U_INT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n
F6	Date/Time (in second since 2000)	IEEE_FLOAT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n
	Coils temperature (°C)	IEEE_FLOAT32			Tc(°C) = n
F7	Date/Time (in second since 2000)	IEEE_FLOAT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n
	Salinity corrected (PSU)	IEEE_FLOAT32			Sc(PSU) = n
F8	Date/Time (in second since 2000)	IEEE_FLOAT32			Datation depends on file name
	Pressure (dbar)	IEEE_FLOAT32			P(dbar) = n
	Temperature (°C)	IEEE_FLOAT32			T(°C) = n
	Salinity (PSU)	IEEE_FLOAT32			S(PSU) = n
	Coils temperature (°C)	IEEE_FLOAT32			Tc(°C) = n
	Salinity corrected (PSU)	IEEE_FLOAT32			Sc(PSU) = n
F9	Date/Time (in second since 2000)	IEEE_FLOAT32			Datation depends on file name

Pressure (dbar)	IEEE_FLOAT32	P(dbar) = n
Temperature (°C)	IEEE_FLOAT32	T(°C) = n
Salinity corrected (PSU)	IEEE_FLOAT32	Sc(PSU) = n

Example: M95C7S1F0A15.hex (Mission: 95, Cycle: 7, Sensor: SBE41, Data format: F0, Phase: ascent, datation frequency: 1/15)



M95C7S1F0A15.hex
SBE41 - ASCENT

Measure Date/Time	P(dbar)	Temp(°C)	Sal.(PSU)
2023/12/08 21:05:22	297.20000	4.09000	34.48000
	296.50000	4.09000	34.48000
	295.50000	4.09000	34.48000
	294.50000	4.09000	34.48000
	293.50000	4.08500	34.48000
	292.50000	4.08000	34.48000
	291.50000	4.08000	34.47500
	290.40000	4.08000	34.47000
	289.50000	4.08000	34.47000
	288.60000	4.08000	34.47000
	287.50000	4.08000	34.47000
	286.50000	4.07600	34.47000
	285.60000	4.07000	34.47000
	284.50000	4.07000	34.47000
	283.50000	4.07000	34.47000
2023/12/08 21:08:13	282.50000	4.07000	34.46500
	281.50000	4.07000	34.46000
	280.50000	4.06500	34.46000
	279.50000	4.05500	34.46000
	278.50000	4.05000	34.46000
	277.50000	4.05000	34.46000
	276.50000	4.05000	34.45500
	275.40000	4.04400	34.45000
	274.50000	4.03500	34.45000
	273.60000	4.03000	34.45000
	272.60000	4.02600	34.45000
	271.50000	4.02000	34.45000
	270.50000	4.02000	34.45000
	269.50000	4.02000	34.45000
	268.50000	4.02000	34.44400
2023/12/08 21:10:29	267.40000	4.02000	34.44000
	266.40000	4.01400	34.44000

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nke Instrumentation
6, rue Gutenberg
ZI de Kerandré
56700 Hennebont
FRANCE
Phone: +33 (0)2 97 36 10 12

www.nke-instrumentation.com

